Binary Classification of AOI Alarms into True Defects and False Calls

Second check of AOI results with supervised machine learning (Siemens production data)

P2: core code and baseline simulation · test plan · report

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Team 1 · Guided by Prof. Dr.-Ing. Hamidreza Heidari · 3 November 2026 · github.com/bodhi584/aoi-false-call-classification

Yu Han

Where the project stands

P1	20 Oct	Proposal, tech stack, MVP (revised topic 23 Oct)	**done**
P2	3 Nov	Core code + baseline simulation; test plan; report	**this phase**
P3	24 Nov	Feature progress; unit tests and CI; issues board	**next**
P4	8 Dec	Feature progress; full evaluation	**planned**
P5	22 Dec	Demo video, run instructions, packaging	**planned**
Final	12 Jan	Final presentation and report	**planned**

P2 in short

Literature review, 31 sources: on the same Siemens data a second check fails on later production days; the authors name two causes, drift and the threshold method.

Research question sharpened: can retraining and a correctly set threshold keep the second check within the targets over time?

Core code: one-to-one replication of that study; first results reproduce it.

Test plan written; report in docs/literature_report.pdf.

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What we changed since the P1 proposal, and why

	P1 proposal (23 Oct)	P2 (now)	Why
Research question	build a second check after AOI, test it in time order	can retraining and a correctly set threshold keep it within the targets over time?	[8] already shows that a check trained once fails later
Starting point	own design	one-to-one replication of Pfab & Rothering (2025)	published code and run logs allow an exact comparison
Models	logistic regression, random forest, neural network	random forest, balanced random forest as in [8]; others as extension	same models as [8]
Evaluation	day-by-day simulation	split of [8]: test set + 5 later slices; train once vs. retrain	same protocol as [8]
Success criteria	board level: recall ≥ 99 %, specificity ≥ 20 %	joint level: missed defects ≤ 1 %, false calls released ≥ 40 %	targets of [8]; savings per board calculated afterwards
Threshold	set on the last 14 days of training	method of [8], then corrected; next: on recent data	[8] names the threshold method as a cause

Unchanged: Siemens AOI data [5] · input → system → output · tools (Python, scikit-learn) · milestones

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Melisa Subasi

AOI false calls are a real and current problem

≈ 99 %

of the solder joints flagged by AOI are false calls

Each flagged board is checked again by hand. Source: Pfab & Rothering (2025); data: Pfab et al. (2024), one Siemens SMT line, 132 days

Zero-defect manufacturing needs detection [1]

AOI checks every solder joint with a camera and fixed limits [2]; machine learning at the inspection step is an active field [3].

Manual workload [5]

Every flagged board goes to the manual inspection station; many are good boards.

Siemens and NVIDIA, Erlangen, Oct 2026 [4]

A second inspection step decides: real defect or false call. Target: 40 % fewer false calls (a target, not a result).

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Meron Hagos

A second check with machine learning works on a test set

Study	Data	Classifiers	Result
Thielen et al. 2020 [6]	not public	kNN, neural network, random forest	false calls recognised; RF and NN best on large data
Albayari et al. 2026 [7]	Siemens AG	XGBoost, RF, DNN, kNN	XGBoost best balance of recall and precision
Pfab & Rothering 2025 [8]	Siemens, public	kNN, RF, balanced RF, XGBoost, AutoML	RF and balanced RF meet the targets on the test set

Which classifiers? [9]–[14]

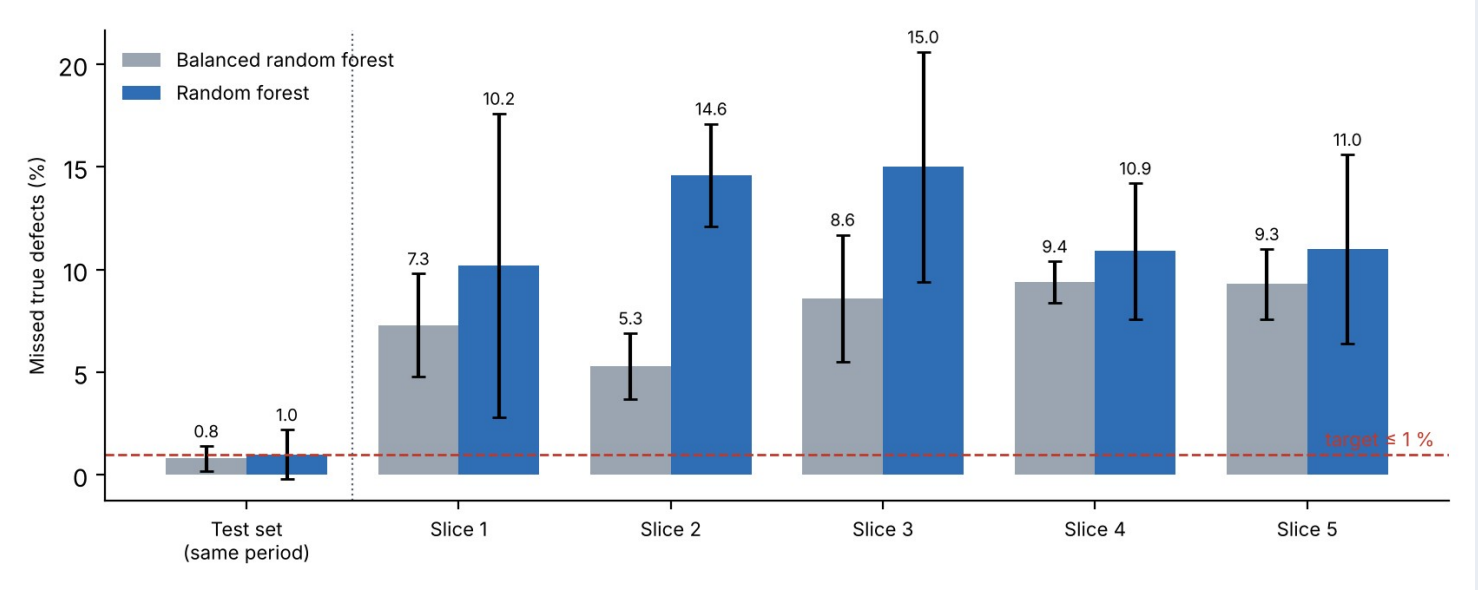
On tabular data, tree ensembles (random forest, gradient boosting) are the strongest family in five benchmark studies.

But: tested on data from the same period

None of these studies keeps the model running on later production days, except [8].

Alejandro Rivas

… but fails on later production days: drift



Why? Drift [22]–[28]

Pfab & Rothering: on target on the test set, 5–15 % missed defects on all five later slices.

Drift = the data change over time; a model trained on old data no longer fits [22]–[24].

Same effect in manufacturing: welding, semiconductor etch, sensors [25]–[27].

The paper also names an unstable threshold method.

Yu Han

Chain of evidence

Fails over time

train once → 5–15 % missed defects later [8]

Two causes

drift [22]–[28] and an unstable threshold method [8]

Remedies

periodic retraining [23], [24], [28]; re-set the threshold when class shares change [21]

Works elsewhere

fraud detection, <1 % positives [30]; semiconductor materials line [31]

Open for AOI

not tested for the second check of AOI results

Our research question

Can periodic retraining and a correctly set threshold keep the second check of AOI results within ≤ 1 % missed true defects and ≥ 40 % fewer manual checks on later production days?

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Input → system → output and what counts as success

INPUT

AOI alarm: AOI measurements of each flagged solder joint

SYSTEM

Binary classifier after the AOI; retrained and threshold re-set during production

OUTPUT

True defect → manual inspection · false call → released

Requirements on the later production periods (joint level, as in [8]; positive class = true defect)

≤ 1 % true defects released (slip = 1 − recall): quality first

≥ 40 % false calls released (volume reduction = specificity): manual checks saved

Evaluated over 5 later slices and 10 random seeds (mean ± SD). Board-level savings for the line are calculated from the same results.

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Replication of Pfab & Rothering (2025): one factor at a time

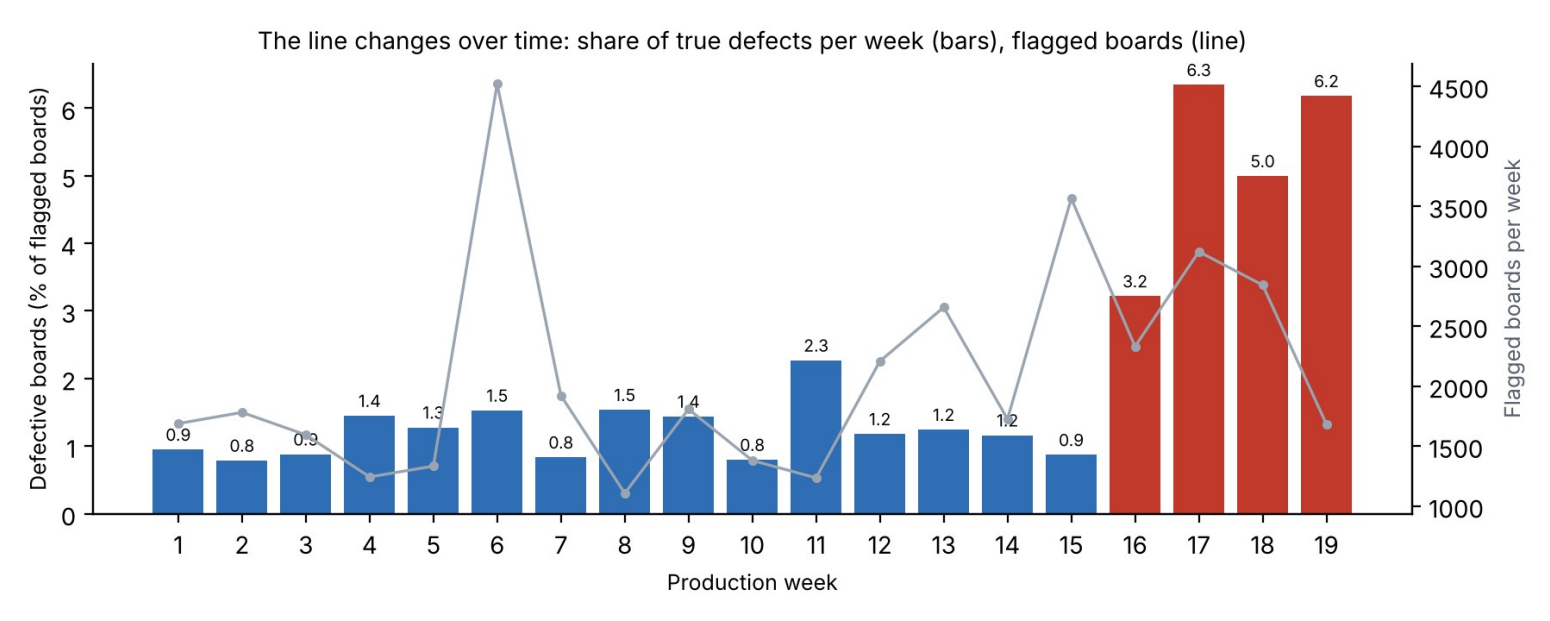
	(a) Paper exactly	(b) Train once, threshold re-estimated	(c) Retrain
Data split	same as [8]	same	same
Models, hyper-parameters, seeds	RF, balanced RF; logged values; 10 seeds	same	same
Threshold	value logged in [8]	their method, applied to the final model	same method, new data
Training	once	once	before each of the 5 later slices
Purpose	reproduce [8]	(a) vs (b) = effect of the threshold	(b) vs (c) = effect of retraining

Metrics on joint level, as in [8]: missed true defects (slip = 1 − recall) and false calls released (volume reduction = specificity). Assumption: manual-inspection labels are available for all joints (upper bound for retraining).

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Alejandro Rivas

The line changes over time



Facts (our analysis)

440,274 flagged joints, 39,742 boards, 132 days

97.8 % of flagged boards are false calls

Share of defective boards: 0.8–2.3 % per week, then 3.2–6.3 % in the last 4 weeks

Drift test: early vs. late days separable, AUROC 0.974

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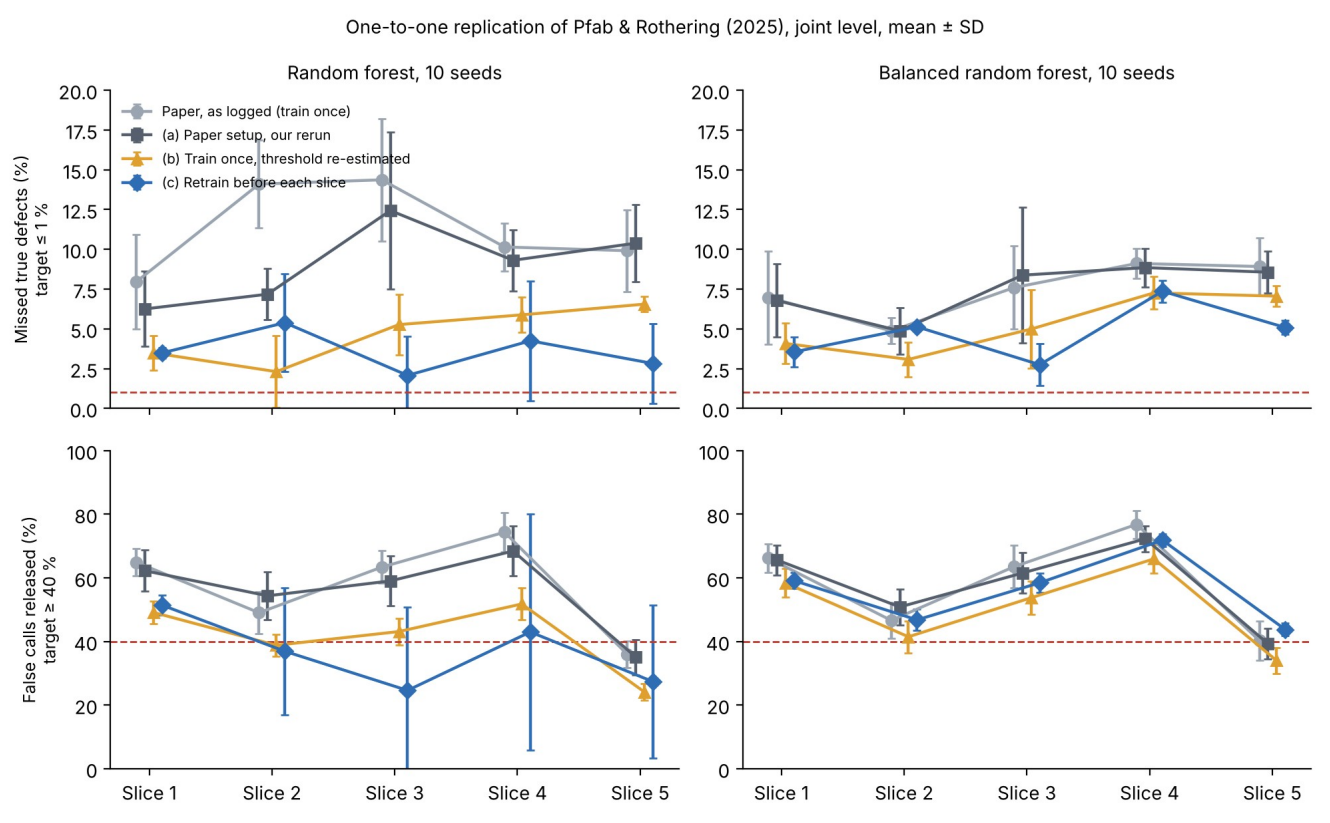
What runs in the repository

Part	File	What it does
Data	aoi_classification/data.py	load the Siemens data, boards, time order
Classifier + threshold	aoi_classification/model.py	train, score, threshold for 99 % recall
Baseline simulation	aoi_classification/simulation.py	day by day: train on earlier days only
Replication	scripts/replicate_pfab2025.py	one-to-one repeat of [8], train once vs. retrain
Classifier screening	scripts/compare_classifiers.py	13 classifiers, 8 families, same split
Tests + CI	tests/, GitHub Actions	10 tests on Windows, macOS, Linux

Tools: Python, pandas, scikit-learn, imbalanced-learn, matplotlib, Jupyter · Data: Mendeley (CC BY-NC 3.0), not stored in the repo

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Replication works; retraining helps a little, target not reached yet



Reproduction ✔

Balanced RF: 7.5 % missed defects, as in [8].

Retraining: small gain

Balanced RF: 5.3 → 4.8 % missed, 50.7 → 56.0 % released.

RF: 4.7 → 3.6 % missed, but fewer released.

Threshold matters most

Correct threshold alone: 7.5–9.1 → 4.7–5.3 % missed.

No variant reaches ≤ 1 % yet.

Melisa Subasi

How we check the code and the numbers

Group	Examples	Status
Data (D1–D7)	labels 0/1, no missing values, time order, checksum	4 exist, rest P3
Classifier + threshold (C1–C7)	threshold rule, all models run, same seed = same result	4 exist, rest P3
Simulation (S1–S4)	model only knows earlier days	1 exists, rest P3
Replication (R1–R4)	same split as [8]; (a) reproduces [8]; retraining uses only earlier data; 10 seeds	new
KPIs (K1–K5)	missed defects ≤ 1 %, false calls released ≥ 40 %; every number from results/	open

Unit tests run on every push (CI, three operating systems). A KPI that is not reached is a result, not a failed test.

Meron Hagos

Towards P3 (24 Nov)

Retraining frequency

daily, every 3 days, weekly, per slice: how far do missed defects drop? [31]

Threshold on recent data

set the threshold on the latest days, not on all history [21]

More model families

gradient boosting, neural network, logistic regression in the same setup

Simulation tool

one screen: choose the retraining frequency, see missed defects and saved checks

KEY REFERENCES

Full list (31 sources) in docs/literature_report.pdf

[5] Pfab, K. et al. (2024). Data of automated optical inspection of surface-mounted technology electronic production. Data in Brief 53, 110110.

[8] Pfab, K., Rothering, M. (2025). Towards improved research methodologies for industrial AI: a case study of false call reduction. arXiv:2506.14521.

[22] Moreno-Torres, J. G. et al. (2012). A unifying view on dataset shift in classification. Pattern Recognition 45, 521–530.

[23] Gama, J. et al. (2014). A survey on concept drift adaptation. ACM Computing Surveys 46(4), 44.

[24] Lu, J. et al. (2019). Learning under concept drift: a review. IEEE TKDE 31(12), 2346–2363.

[30] Dal Pozzolo, A. et al. (2018). Credit card fraud detection: a realistic modeling and a novel learning strategy. IEEE TNNLS 29(8), 3784–3797.

[31] Gao, M. et al. (2026). Robust and reliable AI for predictive quality in semiconductor materials manufacturing with MLOps and uncertainty quantification. arXiv:2605.07752.

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Repository

github.com/bodhi584/aoi-false-call-classification

Thank you

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